

DSL for Table-Top Role-Playing Games (TTRPG) in Python

S.Lott

<https://fosstodon.org/@slott56>
<https://github.com/slott56>

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TTRPG? DSL?

Wait, what?

DSL for
Table-Top
Role-Playing
Games
(TTRPG)

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What's this
about?

Big Lesson Up
Front

Starting
example

Bigger
example

Conclusion

- A Domain-Specific Language (DSL) can be **very** hard to design.
- A Table-Top Role-Playing Game (TTRPG) is a complicated problem domain. (And fun. And low-risk.)
- Designing a DSL for a TTRPG can provide insight into DSL design using Python.

We'll look at two examples.

DSL's Can Be Complicated

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Important

Python may be all you need.

Start there.

Avoid writing a parser.

Dice-Rolling Language

3d6+2

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This is an easy place to start.

`3d6 + 2` is a syntax for rolling polydedral dice. In this case 3 6-sided die; then adding 2.

D&D used platonic solids: d4, d6, d8, d12, d20.

Other games added a d10 (which isn't a platonic solid)

We *can* parse with a regular expression. Or...

In Python

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```
weapon_damage = 3 * D6 + 2
```

```
characteristic = (4 * D6).keep(3)
character = [
    characteristic.roll() for _ in range(6)
]
```

Looks good to me. $3 * D6 + 2$ vs. $3d6 + 2$

Cost: one extra $*$.

How does that work?

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```
class Die:
    def __init__(self, d: int) -> None:
        ...
    def __rmul__(self, n: int) -> "Die":
        ...
    def __add__(self, n: int) -> "Die":
        ...
    def keep(self, k: int) -> "Die":
        ...

D6 = Die(6)
```

The implementation isn't *too* complicated.

OpenD6 Magic Spell Language

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The OpenD6 Fantasy rules define a bunch of magical spells.

There are many pages of rules.

Some aids for doing spell definitions:

- An “Adjusting and Readjusting” admonition.
- A “Spell Design” sheet.
- *These words*: “A calculator might also help.” *Really.*

```
Spell(  
  name="Push",  
  skill="Apportation",  
  notes="Even the most subtle of things...",  
  effect=TimeEffect("sends a small object into the future", "10 min"),  
  duration=DurationAspect(measure="10 minutes"),  
  range=RangeAspect(measure="touch"),  
  casting_time=CastingTimeAspect(measure="1 round"),  
  speed=SpeedAspect.based_on(("range",), ""),  
  other_aspects={  
    "gestures": GesturesAspect("Wave one hand ...", "simple"),  
    "incantations": IncantationsAspect(  
      "Where did it go?", "sentence"  
    ),  
  },  
  other_conditions=[  
    ArcaneKnowledgeAspect(0, "time"),  
  ],  
)
```


That looks acceptable

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Most of the spell details look like this.

```
Spell(  
    ...  
    duration=DurationAspect(measure="10 minutes"),  
    ...  
)
```

The *label = value* is similiary to TOML, JSON, etc.

We can use 2*D as a formal sublanguage where appropriate.

Summary

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Don't invent a DSL from scratch.

Start with Python object definitions as your declarative language.

Don't write a parser unless there's no alternative.

Link to Writing Tools Repository

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<https://github.com/slott56/writing-tools>

Link to OpenD6 Tools Repository

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